The proposed project seems to be a protection hardware for safety critical power systems. The first thing that hit me was which **sensors** to use for this particular application.

I also chose to use mechanical relays for controlling the live wire connection. Also for the **configuration** interface i chose **UART**, which can be easily done using serial terminal like **PuTTY** from a PC.

Luckily i had worked on similar project during my undergrad so had a couple of sensors in mind. I choose to use **ACS712-30A** module for current sensing and **ZMPT101B** module for voltage respectively. Both these sensors can be easily prototyped on arduino and therefore was first choice.

Moving on I needed to know whether these sensors can be put together to come up with the expected protection hardware with a **baremetal** setup on **arduino nano**. If everything works as expected we could upgrade the system to more sophisticated architecture with **RTOS** on a **esp32** device.

The baremetal setup uses a **superloop** and arduino **millis()** function to initiate activities (**sampling**, **protection** and **serial**). On a RTOS setup the activities are taken care of by **tasks** (xTaskCreate) that independently runs round-robin fashion synchonized with **mutexes** (stateMutex and serialMutex).

Unfortunately I don't have the **ACS712-30A** and **ZMPT101B** either at the moment and simulating the complete system on simulation platforms like **wokwi** don't seem to have characterized library for these sensors.

Proteus could've been an option but I'm on **Linux** which adds another dilemma in the process. So simulated the system with **fake** ADC samples on athe arduino nano baremetal setup for testing purpose. Included the snapshots along with source code (please checkout **baremetal** and **rtos** branch)